LESSON - 14

UNITS DIMENSIONS / ERROR ANALYSIS

Physical Quantities (Mesured Values)

i. Fundamental
ii. Derived
iii. Supplementary

Units : Measurement of Physical Quantity is Expressed in terms of internationally accepted Standard Called Units.

Fundamental Quantities :

(i)	Mass	: Kg
(ii)	Length	: m
(iii)	Time	: S
(iv)	Electric Current	: A
(v)	Thermodynamic Temp	: K
(vi)	Amount of Substance	: mol
(vii)	Luminous intensity	: Cd.

Derived : Speed Density

$\frac{m}{Sec}$ $\frac{Kg}{m \times m \times m}$

Supplementary : Only Units NO Dimensions..

(i) Plan Angle : Radians.
(ii) Solid Angle : Steradians..

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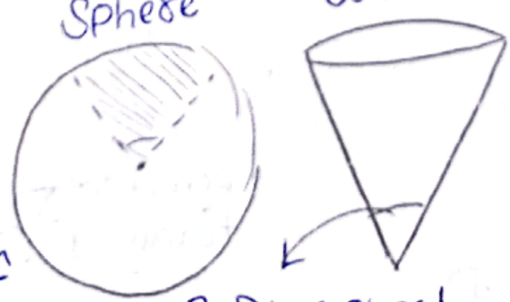
$\boxed{\text{Area Circle Patch } 2\pi R^2(1-\cos\theta)}$

Solid Angle :

Sphere Cone



3 Dimensional Angle...

(Solid Angle)

$$\boxed{\Omega = \frac{\text{Area of the Patch}}{R^2}}$$

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$\Omega = \frac{2\pi R^2(1-\cos\theta)}{R^2}$

$\Omega = 2\pi(1-\cos\theta)$

Linear charge density : $\lambda = \frac{q}{l}$

Intensity = $\frac{\text{Energy}}{\text{Area} \times \text{time}}$

- Magnitude of P.Q = (Numerical Value) × Unit
  Physical Quantity

- Numerical Value $\propto \frac{1}{\text{Unit}}$

$B = \frac{F}{qv}$ $E = \frac{F}{q}$

$n_1 u_1 = n_2 u_2 = \text{Constant}$

Example.

10m = 1000cm.

Viscous force $\boxed{F = \eta A \frac{dv}{dy}}$

$\eta = [M L^{-1} T^{-1}]$

	SI Unit	D.F
Velocity	m/s	$M^0 L T^{-1}$
Momentum	Kgm/s	$M^1 L T^{-1}$
Acceleration	m/s^2	$M^0 L T^{-2}$
Force	Kgm/s^2	$M L T^{-2}$
Pressure	$\frac{Kg}{ms^2}$	$M L^{-1} T^{-2}$

Form: $[M^a L^b T^c]$

Mechanics

Force = $M L T^{-2}$

Energy = $M L^2 T^{-2}$

Current

Electricity

$i = \frac{dq}{dt} = \frac{c}{sec} = \text{Amp/sec}$

$\boxed{\frac{\text{Joule}}{\text{coul}} = \text{Volt}}$

$\boxed{\text{Pressure} = \frac{F}{A}}$

$\boxed{KE = \frac{3}{2}KT}$

	SI Unit	D.F
Work	Joule	$M L^2 T^{-2}$
Energy	Joule	
Kinetic	Joule	
Power	$\frac{\text{Joule}}{\text{sec}}$	$M L^2 T^{-3}$

Permittivity $\Rightarrow [M^{-1} L^{-3} T^4 A^2]$

Farad = $M^{-1} L^{-2} T^2 Q^2$

Permittivity

Permeability : $M L T^{-2} A^{-2}$ [∵ A is Current]

Inductance : $ML^2T^{-2}A^{-2}$

	SI Units	D.F
Moment of inertia	Kgm^2	ML^2
Surface Tension	$\frac{N}{m} = \frac{Kg}{s^2}$	MT^{-2}
Torque	N-m	ML^2T^{-2}
Stress/Youngs Modul	$\frac{N}{m^2}$	$ML^{-1}T^{-2}$

$\vec{E} = \frac{F}{q}$

$\vec{M}$ (B) $= \frac{F}{qV}$

$\frac{\text{Work Done}}{\text{Charge}}$ ...

Voltage (V) : $\frac{W}{Q} = \frac{\text{Joule}}{\text{Coulmb}}$

Charge $\frac{dQ}{dt} = i$

$dq = i\,dt$

Hence

$[M^0L^0T^1A^1]$

$T = \frac{F}{l}$

This is a Property.

Longitudinal Stress :

F ← → F

	SI Units	Dimensional Formula.
Strain	—	—
Refractive index	—	—
Universal gas constant	$\frac{\text{Joule}}{\text{Mol.K}}$	$[ML^2T^{-2}Mol^{-1}K^{-1}]$
(K) Boltzmann Constant	$\frac{\text{Joule}}{\text{Kelvin}}$	$[ML^2T^{-2}K^{-1}]$
Plank constant	Joulesec	ML^2T^{-1}
Gravitational cons	$\frac{Nm^2}{Kg^2}$	$M^{-1}L^3T^{-2}$
Resistance	—	$ML^{-1}L^{-2}T[\text{Coulomb}]^{-2}$
Magnetic field	Tesla	$F = Bil$ $\boxed{B = \frac{F}{il}}$ $B = \frac{N}{Ampm} = [MT^{-2}A^{-1}]$
Electric Field	$\left(\frac{N}{C}\right)$	$\left[\frac{MLT^{-2}}{AT}\right]$

~~Torque.~~

$[MLT^{-3}A^{-1}]$ $[M^aL^bT^cA^d] \rightarrow$ Current Electricity...

Coulomb $= [A][T]$

→ $E_0 \rightarrow M^{-1}L^{-3}T^4A^2$

→ C (Capacitor) $= \frac{1}{R}$ $[M^{-1}L^{-2}T^4A^2]$

→ Resi...

[Thermodynamic]

$PV = nRT$

$R = \frac{PV}{nT}$

$= \frac{\left(\frac{N}{m^2}\right)m^3}{mol \cdot K}$

$R = \frac{\text{Joule}}{\text{Mol.K}}$

Specific Heat

$Q = ms\Delta Q$

$S = \frac{\text{Joule}}{\text{KgK}}$

$S = L^2T^{-2}K^{-1}$

Latern Heat

$Q = mL$

$L = [L^2T^{-2}]$

Derivision:

$$K = \frac{R}{N_A} = \frac{\frac{\text{Joule}}{\text{Mol K}}}{\frac{1}{\text{Mol}}} = \frac{\text{Joule}}{K}$$

$$\boxed{U = \frac{3}{2} KT}$$

$K \rightarrow \frac{\text{Joule}}{\text{Kelvin}}$

$K = [ML^2T^{-2}K^{-1}]$

Energy density $= \frac{\text{Energy}}{\text{Vol.}}$

** Stephens Boltzeman Constant (a)

$$\boxed{P = aAeT^4}$$

$a = 5.67 \times 10^{-8} \frac{\text{watt}}{M^2K^4}$

$a = \left[\frac{ML^2T^{-3}}{L^2K^4}\right]$

$a = [MT^{-3}K^{-4}]$

Gravitation.

$F = \frac{GM_1M_2}{r^2}$

$G = \frac{Nm^2}{Kg^2}$

$= \frac{MLT^{-2}L^2}{M^2} \quad M^{-1}L^3T^{-2}$

** Wienn's Displacement Constant: $(\lambda)(T)$

Electric Potential $\frac{\text{Joule}}{\text{coul}}$ = Volt.

Magnetic Permeability:

Coefficient of Viscosity: (η) ⟶ Property of fluid.

Viscosity: (Strok's Law) $\boxed{f_v = 6\pi\eta r v.}$

$f_v = \eta A \underbrace{\left(\frac{dv}{dy}\right)}$ ⟶ Velocity Gradiant...

$N = \eta \cdot (m^2)\left(\frac{m/s}{m}\right)$

$\eta = \frac{N \cdot sec}{m^2} = \frac{Kgm}{s^2} \cdot \frac{sec}{m^2}$

$$\boxed{\eta = \frac{Kg}{M \cdot sec}}$$

$\eta = ML^{-1}T^{-1}$

Viscosity:

→ SI Unit: $\frac{Kg}{M\,sec}$ = 1 Poisuelle

→ CGS Unit:

$\frac{gm}{cm \cdot sec}$ = 1 Poise

Relation b/w Poisuelle/Poise:

$\boxed{1 \text{ Poisuelle} = 10 \text{ Poise}}$ ***

1. Which have Different Dimensions:

× (A) Pressure, Young's Modulus, Stress

× (B) Emf, Potential Difference, Electric Potential

× (C) Heat, Work Done;

(D) Dipole Moment, Electric . . . .

$\boxed{\text{Rule 1:}}$

(Principle of Homogeneity of Dimensions)

Unit 1 + Unit 2 = Unit 3

Unit 1 − Unit 2 = Unit 3

Dimension of Unit 1 = Unit 2 = Unit 3.

A+B = C

A−B = C

A, B, C Should have Same Units.

Example: $X = A + BC + \frac{D}{E} + \frac{P^2}{q} + -\sqrt{\frac{r}{s}}$

m m m m m m.

If B = Kg

Find C. Use Homogeneity $\boxed{C = \frac{m}{Kg}}$

2. A+B = 2Kg Units of A : Kg

B : Kg

3. A+B×C = 2Kg Units of A : Kg if $B = m^3/kg$

C : $\frac{kg^2}{m^3}$

4. $\boxed{V = a + bt + \frac{c}{d+t}}$

Dimensions of abcd.

$a \rightarrow m/s$

$b \rightarrow m/s^2$

$c \rightarrow m$

$d \rightarrow Sec.$

Homogen~

$\boxed{\frac{c}{Sec} = \frac{M}{Sec}}$

5. $\left(P - \frac{a}{V^2}\right)(V - b) = nRT \longrightarrow$ Vaanderwalls forces.

Find Dimension

$a \rightarrow (\ ).$

$b \rightarrow m^3 = [L^3].$

$a = Nm^4$

$a = \frac{Kgm}{S^2} . m^4$

$a = [ML^5T^{-2}].$

Rule 2:

- Trigonometric Ratios, Exponetial, Logarithmic Are Dimension Less.

Example:

: $\sin\left(\frac{a^2t}{b}\right)$

$\boxed{\frac{a^2}{b} = \frac{1}{Sec}}$

Errors & Measurements :

1. Systamatic Error : The Exact cause of Error is Known to us (Rectified)
2. Random Error : Exact cause of the Errors is not Known...

$$\text{True Value} = \frac{\text{Mean}}{\text{No. of Things}}$$

$$\boxed{\text{Error} = T.V - \text{Measured Value}}$$

$$\text{Mean Absolute Error} = \frac{|\Delta_1| + |\Delta_2| + |\Delta_3| + \dots + |\Delta_n|}{n}$$

Propagation of Errors :

1. $\boxed{x = a+b}$

$a = 7.4 \pm 0.2$

$b = 3.2 \pm 0.1$

$x = (10.6 \pm 0.3)$

$\pm \Delta x = \pm(\Delta a + \Delta b)$

2. $\boxed{x = a-b}$

$a = 7.4 \pm 0.2$

$b = (3.2 \pm 0.1)$

$x = (4.2 \pm 0.3)$

$\pm \Delta x = \pm(\Delta a + \Delta b)$

3. $x = ab$

$\pm \frac{\Delta x}{x} = \pm\left[\frac{\Delta a}{a} \pm \frac{\Delta b}{b}\right]$

Not Possiable

4. $x = \frac{a}{b}$

$\pm \frac{\Delta x}{x} = \pm\left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$

5. $x = K \cdot \frac{a^p b^q}{c^r}$

K = constant

$$\boxed{\pm\frac{\Delta x}{x} = \pm\left(p\frac{\Delta a}{a} + q\frac{\Delta b}{b} + r\frac{\Delta c}{c}\right)}$$

Homogeneity Principle.

LHS = RHS

i. Rounding Off.

ii. Significant figures...